ABDULAKEEM ADEBAYO

Full-Stack Developer | Data Scientist | AI Automation Engineer | Product Analyst | Python | React | Next.js | Node.js

Contact Information

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- **Portfolio**:abdulakeem-adebayo.vercel.app

Professional Summary

Versatile Full-Stack Developer, Data Scientist and AI Automation Engineer who designs and ships complete products: real-time web platforms, cross-platform mobile apps, REST/WebSocket APIs, machine learning pipelines and LLM-powered automations. Highlights include a 13-game multiplayer platform, a finance app with 347 automated tests, a role-secured medical records system, a football prediction model retrained weekly on ~9,000 matches, and AI agents that read emails and run multi-step workflows. In industry, my analysis and models lifted retention by 15% and engagement by 25%.

Work Experience

Freelance AI Automation, Data & Full-Stack Developer
Web & Mobile Development, ML, LLM Integration | Remote
2023 – Present

- Built **VersusX**, a real-time multiplayer games platform (**Node.js, WebSockets, SQLite, PWA, Three.js**) with 13 games including full **FIDE chess** (perft-tested move generator), draughts, Ludo, Whot and 8-ball, plus **Glicko-2 ratings**, matchmaking, game clocks, reconnects and server-side bots. Deployed on **Vercel + a VPS**.
- Built **Finlo**, a cross-platform personal finance app for **Android, iOS and web from one codebase** (**Expo, React Native, TypeScript, SQLite**): multi-currency money engine on integer minor units, time-aware budgets, receipt OCR and read-only Gmail import, and PDF/Excel/CSV reports, covered by **347 automated tests**.
- Built **Caresync EMR**, an electronic medical records platform (**pnpm monorepo: Next.js 15 web, Expo mobile, Supabase**) with **7 roles enforced by Row-Level Security** (doctor, nurse, lab scientist, pharmacist, patient…), covering patients, appointments, consultations, lab reports, prescriptions and drug inventory.
- Built **ShiftPilot**, a shift-scheduling app for agency workers (**Expo Router, TypeScript, Supabase/Postgres with Row-Level Security, realtime**): conflict detection in SQL, a **pg_cron "leave-by" engine** that uses Google Maps live traffic to send push alerts, and an **email webhook where Claude extracts shift details** into a review queue.

- Built a **football shots prediction model** (**Python, XGBoost, Poisson thresholds, SQLite, Streamlit, GitHub Actions**) trained on about **9,000 matches across 8 competitions**: leakage-safe rolling features, opponent defence z-scores, time-based holdout and drift checks, retrained automatically every week.
- Built **Sports Bet Claude**, a sports analytics engine (**Python, Flask, Playwright, Docker**) that collects form, xG and head-to-head data across 20+ competitions and 4 sports, scores picks with **sample-size-shrunk hit rates**, grades every pick against real results, and **re-calibrates its own confidence** each week.
- Engineered a **15-component Python ML ensemble trading system** combining **LSTM, Transformer, XGBoost and Random Forest** models with **FinBERT** news sentiment analysis and a real-time web monitoring dashboard.
- Built **PROWL**, an AI crypto-research agent with a live dashboard (**React 19, Vite, Tailwind, Supabase, Python backend**): a **multi-agent pipeline** where Scout finds trending narratives on X/Telegram, Guard runs **17 contract-safety checks** via the GoPlus API, and an execution agent acts on approved tokens.
- Built modular **browser automation systems** with **Python and Playwright**, plus **Flask + Socket.IO dashboards** for real-time control and status tracking of parallel automated browser sessions.
- Designed and contributed **adversarial evaluation tasks for frontier LLM benchmarking** projects, probing model capability limits through binary recovery scenarios and domain-specific knowledge tasks.

Technotrend Platforms Nigeria Ltd.
Full-Stack Web & Backend Developer / Data Specialist / Product Analyst
2023 - 2025

- Coordinated **VAS (Value-Added Services)** product implementations including **Miniloan** and **Call Collect**, managing communication between client stakeholders and third-party providers using **Voice Event Manager (VEM)** for IVR-based delivery.
- Architected and developed responsive web interfaces and **admin dashboards** (**HTML5, CSS3, JavaScript, Node.js, Python**) for managing VAS subscriptions, user onboarding and automated reporting workflows.
- Monitored and reported on VAS sales funnel performance across **IVR, USSD, SMS and Web**, tracking **KPIs** such as **CTR**, conversion rate, **ARPU** and churn to find optimisation opportunities.
- Facilitated implementation of **fraud detection** mechanisms and **double opt-in validation** workflows, aligning client requirements with provider capabilities for regulatory compliance.
- Conducted data analysis that identified key user behaviours, leading to a **15% increase in user retention** and a **20% boost in website performance**.
- Implemented **A/B testing**, optimising features and improving customer satisfaction by **18%** based on statistical results.
- Developed and deployed **machine learning models** for personalised content recommendations, resulting in a **25% increase in user engagement**.
- Created interactive reports and dashboards using **Tableau, Excel and Power BI**, accelerating decisions and reducing report generation time by **30%**.

Extern | Beats by Dre.
Consumer Insights and Market Intelligence (Remote Extern)
2024

- Researched **macro consumer trends among Gen Z** and benchmarked brand sentiment against key competitors.
- Developed and presented a **data-driven strategy** for connecting with Gen Z, using surveys, interviews and market analysis.

Azubi Africa
Data Analytics Professional
2022 - 2023

- Led multiple data analytics projects, improving team efficiency by **20%** through streamlined communication and task allocation.
- Developed machine learning models that enhanced data analysis, leading to a **10% improvement in prediction accuracy** and better resource allocation.

Earlier Experience

- **O'dua Telecoms, Onet** – *Technology Intern* (2017 - 2018): Provided technical support and troubleshooting, improving system uptime by **15%**.
- **Africa Internet Holdings (JUMIA)** – *Quality Control* (2015): Monitored and enforced quality standards across operations, reducing the defective product rate by **10%**.
- **Africa Internet Holdings (KAYMU)** – *Content Manager* (2013): Managed and optimised site content, leading to a **12% increase in site traffic** and improved SEO performance.

Education

Obafemi Awolowo University
Bachelor of Computer Science with Economics
2014 - 2021

Certifications

- **AWS Cloud Practitioner** (2022) (**Certified**)
- **IIM Certified Data Protection Officer (CDPO), issued under NDPC authority** (2026) (**Certified**)
- **ALX AiCE AI Career Essentials** (2024) (**Certified**)
- **Google Data Analytics Professional** (2022) (**Certified**)
- **Azubi Africa Data Analytics Professional** (2022 - 2023) (**Certified**)
- **McKinsey Forward Thinking** (2022 - 2023) (**Certified**)
- **Python for Financial Analysis and Algorithm Trading** (2020)

Skills

Technical Skills

- **Frontend:** React, Next.js 15, TypeScript, JavaScript (ES2023), HTML5, CSS3, Tailwind CSS, Framer Motion, Bootstrap
- **Backend:** Node.js, NestJS (QA contributor), Python (Flask, FastAPI, Django), REST APIs, WebSockets / Socket.IO
- **Databases:** PostgreSQL, Supabase (Row-Level Security, Edge Functions, pg_cron, realtime), MySQL, SQLite, BigQuery, Prisma
- **Mobile:** React Native, Expo (EAS builds, Expo Router), react-native-web, NativeWind, Zustand, MMKV, SQLite
- **Python data stack:** pandas, NumPy, scikit-learn, XGBoost, deep learning (LSTM, Transformer), Jupyter

- **Machine learning:** classification, regression, time-series forecasting, ensembles (LSTM, Transformer, XGBoost, Random Forest), Poisson models
- **SQL:** PostgreSQL, MySQL, BigQuery, SQLite
- **BI & Visualisation:** Power BI, Tableau, Excel (XLOOKUP, Pivot Tables, VBA, simulations), Streamlit, Matplotlib/Seaborn
- **LLM / AI APIs:** Claude (Anthropic), Gemini, function calling / tool use, structured extraction
- **Browser automation:** Playwright, Selenium, multi-instance orchestration, Flask + Socket.IO control dashboards
- **Telecom VAS:** IVR, USSD, SMS, Voice Event Manager, prepaid charging and mediation
- **Product & KPI analytics:** CTR, CVR, ARPU, churn, retention, funnel analysis
- **MQL4/MQL5:** Forex and crypto trading bots, backtesting
- **DevOps & Cloud:** Git/GitHub, GitHub Actions, Docker, Vercel, VPS deployment (Vultr), AWS (Cloud Practitioner)
- **Soft skills:** Analytical Thinking, Problem Solving, Ownership of end-to-end delivery, Clear technical documentation, Fast Learner, Time Management
- **Languages:** English (Native), French (Basics), Spanish (Basics), Arabic (Basics)

Projects/Articles

Personal Projects

- **VersusX** – real-time multiplayer games platform, 13 games (Node.js, WebSockets, Glicko-2, Three.js) – versusx-theta.vercel.app
- **Finlo** – multi-currency budgeting and forecasting app (Expo, React Native, TypeScript, SQLite, Gemini)
- **Caresync EMR** – role-based medical records platform (Next.js 15, Expo, Supabase RLS)
- **ShiftPilot** – shift scheduler with live-traffic leave-by alerts and AI email parsing (Expo, Supabase, Claude)
- **CodeDial** – offline USSD directory and spend tracker for Nigerian networks and banks (React Native, Expo)
- **J2D Online Academy** – teaching portal with student, teacher and admin dashboards
- **Football Shots Model** – XGBoost + Poisson player-shots predictor with weekly auto-retrain (Streamlit)
- **Sports Bet Claude** – self-calibrating sports analytics engine (Python, Flask, Playwright, Docker)
- **ML Ensemble Trading System** – LSTM, Transformer, XGBoost, FinBERT sentiment
- **PROWL** – multi-agent AI crypto-research dashboard (React 19, Supabase, Python) – prowl-agent.xyz
- **Customer Churn ML Prediction** – classification models (published on Medium)
- **Data Analysis on Indian Start-Up Funding** – EDA and Power BI dashboard
- **Bitcoin Price Prediction Application** using LSTM
- **Forex and Crypto Bots** – MQL5 Expert Advisors and a Python SMC backtester
- **Frontier LLM Adversarial Benchmarking** – evaluation task design

Article/Visualizations Links

- **Live projects & code:** Portfolio • VersusX • PROWL • GitHub
- **Customer Churn ML:** Medium article • Power BI • GitHub
- **Indian Start-Up Funding:** Medium article • Power BI